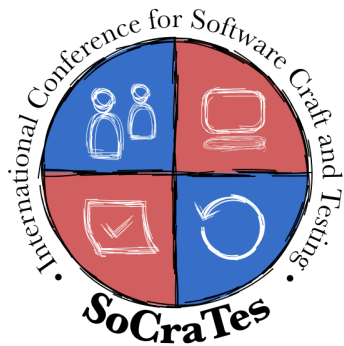


My Journey to Selfhosting & Homeserve

Patrick Drechsler





Motivation

- Privacy / Enshittification
- Learning: networks, operations, IaC, etc.
- Saving money?



Disclaimer

- I am new to selfhosting
- This is my learning experience
- slowly reaching status "Enough to be dangerous" 🐱





Cloud or Homeserver?



Homeserver

Pros

- ✓ full control
- ✓ privacy: all data is yours
- ✓ no vendor lock-in
- ✓ low running costs

Cons

- ✗ initial hardware investment costs
- ✗ maintenance of all infrastructure
- ✗ availability / scaling



Cloud

Pros

- ✓ Dedicated machine OR
- ✓ PaaS / SaaS solutions
- ✓ availability / scaling
- ✓ no hardware investment
- ✓ easier setup

Cons

- ✗ vendor lock-in
- ✗ running costs are difficult to predict
- ✗ trust issues

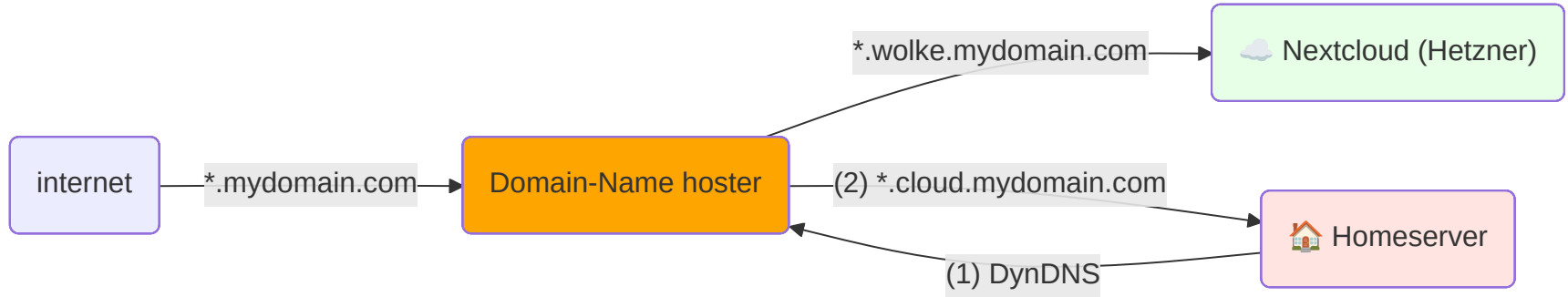


Typical Applications

app	"before"	"after"
 calendar	Google Calendar	 Nextcloud (Hetzner)
 file sharing	Google Drive	 Nextcloud (Hetzner)
 photos	Google Photos	 <u>immich</u>
 document management	private git	 <u>Paperless-NGX</u>
 read-it-later	Pocket (Mozilla)	 <u>Readeck</u>
 cooking recipes	bookmarks, Chefkoch, etc	 <u>Mealie</u> , <u>Tandoor</u>



Domain Hosting (current state)



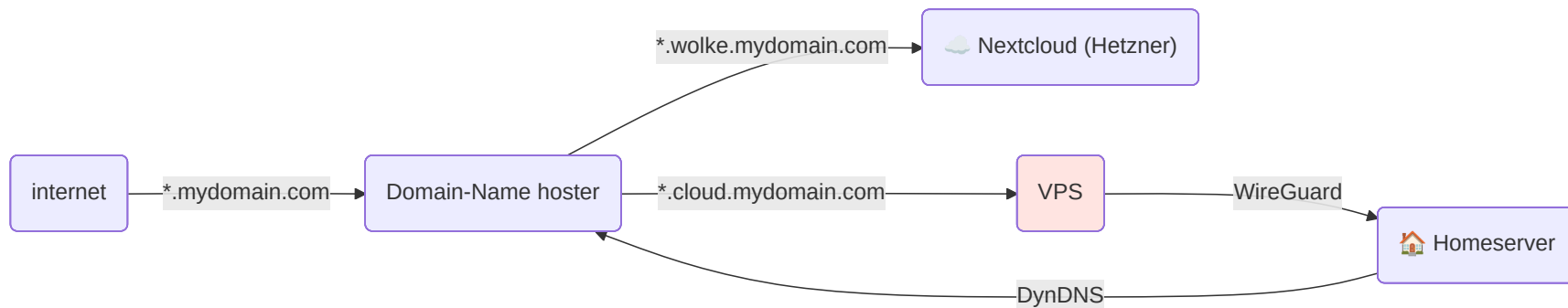
I learned something about

- `CNAME` entries
- `A` entries
- wildcards



Domain Hosting (alternative)

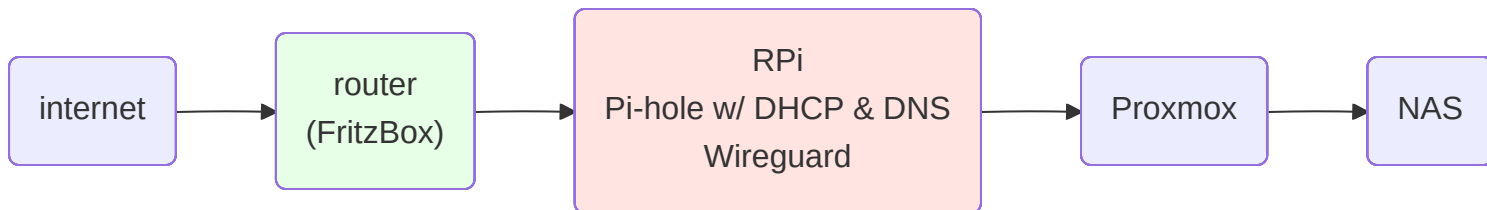
Use a Virtual Private Server (VPS) to access Home server



- on my "needs more research" list, maybe overkill?
- Safer, but might involve extra costs



Home infrastructure: Bird's eye view



router (FritzBox)

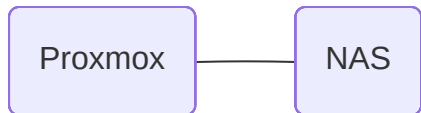
- I try to keep the router as dumb as possible
- disabled DNS & DHCP
- enabled DynDNS
- Let's Encrypt: port 80/443 is open ⚠
- wireguard port is open
- configs are easy to backup & restore

RPi

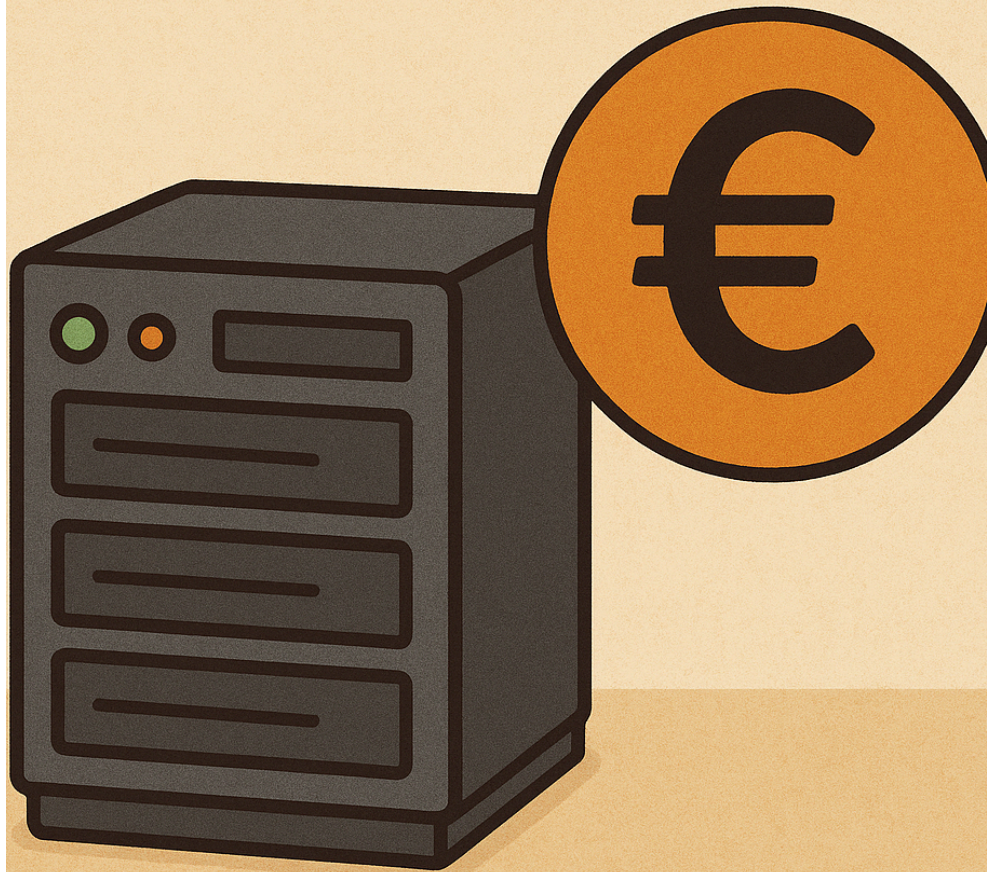
- Raspberry Pi 2 Model B (!) 🕶
- Pi-hole: DNS & DHCP
- WireGuard (via PiVPN)
- configs are easy to backup & restore



Home infrastructure



Hardware: Let's talk about money...





Hardware (1/2): NAS

My old Synology (from 2014)...

- 4 HDDs
- costs (mainly the HDDs): 1000 EUR
- effective storage capacity: 8TB
- Synology's "RAID-5"

YMMV, but you want something solid & tested.





Hardware (2/2): Server

- costs: 320 EUR
- CPU: Intel Alder Lake N95
- RAM: **32GB**
- SSD: **1TB (max 8TB)**



Rufus

run Windows?

Does it have built-in Wi-Fi?

[Click to see full view](#)



DreamQuest Mini PC 195 Processor 32GB +1TB SSD Mini Desktop PC 4K@60Hz/Gigabit Ethernet Suitable for Daily Office Use

[Visit the DreamQuest Store](#)

3.5 ★★★★★ (53)

€319⁰⁰

Or €108.12 per month (3 months)
with 10,01% APR [Select plan](#)

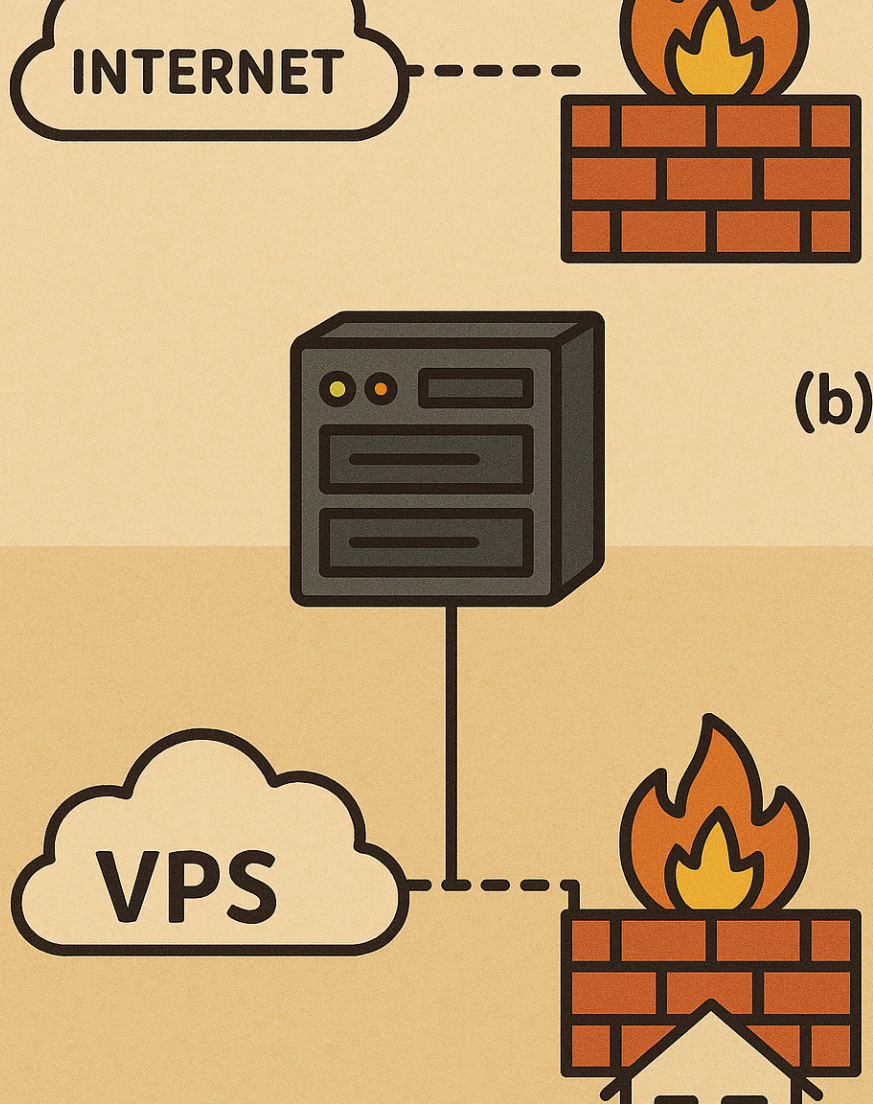
✓prime

FREE Returns

Prices for items sold by Amazon include VAT. Depending on your delivery address, VAT may vary at Checkout. For other items, please see [details](#).

[Want to recycle your product FREE of charge?](#)

Brand	DreamQuest
Operating system	One Size
CPU model	Celeron
CPU speed	3.4 GHz
Cache size	1 TB
Graphics card	Integrated



Accessing Homeserver

- VPN / Wireguard
- Public access (w/ Geoblocking)

Thoughts? 🤔



Why Proxmox VMs?

I am not an expert: The more contained my experiments are, the safer I feel.

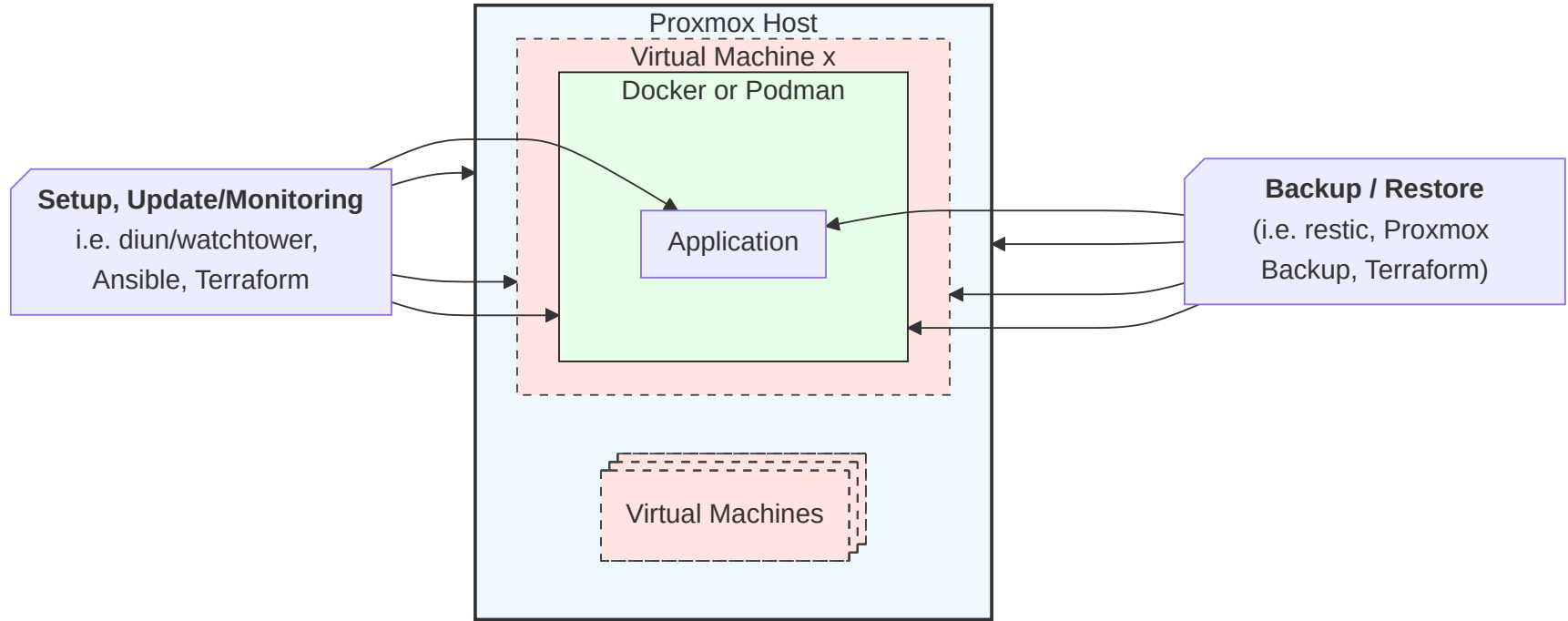
- VMs: **containment** → safety feature
- builtin **backups**
- VMs can be **orchestrated** with
 - **Ansible** (inside the VM)
 - **Terraform** (spinning up the VM)

Alternatives (for pros): kubernetes, nix-os





Server: Overview





Outlook

TODOs

- Setup VMs w/ Terraform
- Add `fail2ban` to public facing `caddy`
- Centralized logging
- Centralized Monitoring
- Research Virtual Private Server (VPS) - esp. costs

We can have a look at details

- Proxmox (live)
- Ansible scripts
- anything else in my network!

Let's talk!





Thank You!

- ✉ patrick.drechsler@mathema.de
- 🐙 <https://github.com/draptik>
- 🌐 <https://draptik.github.io/talks/>
- 💬 <https://floss.social/@drechsler>
- 🦋 <https://bsky.app/profile/drechsler.bsky.social>
- 🔗 <https://www.linkedin.com/in/patrick-drechsler-draptik/>

Slides 🖱️

- QR Code or
- <https://draptik.github.io/2025-09-socrates-at-selfhosting/>

